Abstract 30 Table 1

	$LV_{non\text{-}remod}$ (N=26)	LV_{remod} (N=23)	p-value
EF at 3M (%)	53 ± 7	41 ± 10	<0.001
Stroke Volume (mL)	75 ± 18	73 ± 13	0.9
IS (%)	10.6 ± 11	27 ± 10	<0.001
GRS	23.5 ± 7.6	16.3 ± 7	<0.001
GCS	-15.1 ± 1.6	-12.2 ± 2.7	<0.001
GLS	-12.4 ± 2	-9.4 ± 3	<0.001
Direct-flowi$_{EDV}$	35.3 ± 8.2	20.7 ± 7.3	<0.001
Systolic-KEi$_{EDV}$	9.3 ± 3.4	7.3 ± 2.1	0.03
Residual LV-Volumei$_{EDV}$	22.5 ± 10.1	37.7 ± 17.7	<0.001
In-plane-KEi$_{EDV}$	0.3 ± 0.1	0.4 ± 0.4	0.02

change in intracavitary systolic kinetic energy and flow which were not depicted by stroke volume, suggesting an early interaction between FT, KE and 4D flow parameters at 3M post STEMI in the LV_{remod} group.

Conclusion The interplay between LV mechanistic and flow kinetic at 3 months post-acute MI leading to adverse remodeling at 12 months involves significant alteration of FT,4D flow and systolic KE parameters. These parameters could help for further risk stratification for the development of adverse remodeling at 12 months post-acute MI.

Acknowledgements

31 AORTIC FLOW ABNORMALITIES CAN DIAGNOSE HEART FAILURE WITH PRESERVED EJECTION FRACTION

[1]Zia Mehmood*, [1,2]Hosamadin Assadi, [1,2]Rui Li, [1,2]Bahman Kasmai, [1,2]Gareth Matthews, [1,2]Ciaran Grafton-Clarke, [1]Aureo Sanz-Cepero, [3]Xiaodan Zhao, [3,4,5]Liang Zhong, [6,7]Nay Aung, [1]Kristian Skinner, [1]Charaka Hadinnapola, [8]Peter Swoboda, [9]Andrew J Swift, [1,2]Vassilios S Vassiliou, [10]Christopher Miller, [11]Rob J van der Geest, [6,7]Stephen Peterson, [1,2]Pankaj Garg. [1]*Norfolk and Norwich University Hospitals NHS Foundation Trust, Norwich, Norfolk, UK*; [2]*Department of Cardiovascular and Metabolic Health, Norwich Medical School, University of East Anglia, Norwich, Norfolk, UK*; [3]*National Heart Research Institute Singapore, National Heart Centre Singapore, Singapore*; [4]*Cardiovascular Sciences Academic Clinical Program and Cardiovascular Metabolic Disorder Program, Duke National University of Singapore Medical School, Singapore*; [5]*Department of Biomedical Engineering, National University of Singapore*; [6]*William Harvey Research Institute, NIHR Barts Biomedical Research Centre, Queen Mary University of London, London, UK*; [7]*Barts Heart Centre, St Bartholomew's Hospital, Barts Health NHS Trust, London, UK*; [8]*Division of Biomedical Imaging, Leeds Institute of Cardiovascular and Metabolic Medicine, University of Leeds, Leeds, UK*; [9]*Department of Infection, Immunity and Cardiovascular Disease, University of Sheffield, Sheffield, UK*; [10]*Division of Cardiovascular Sciences, School of Medical Sciences, Faculty of Biology, Medicine and Health, Manchester Academic Health Science Centre, University of Manchester, Manchester, UK*; [11]*Department of Radiology, Division of Image Processing, Leiden University Medical Center, Leiden, The Netherlands*

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Introduction There is growing interest in identifying cardiovascular magnetic resonance (CMR) signatures in ageing due to their relevance to cardiovascular health.[1] It also remains uncertain whether patients with heart failure with preserved ejection fraction (HFpEF) have disruptions in their aortic flow. This study aimed to explore sophisticated indicators of aortic flow disturbances in ageing and in HFpEF.

Materials and Methods This study used two-dimensional phase-contrast CMR data at an orthogonal plane just above the sino-tubular junction. We recruited 10 young healthy controls (HCs), 10 old HCs and 23 patients with HFpEF. We

Abstract 31 Table 1 Aortic flow indices trend across the three groups

	Young HCs (10)	Old HCs (10)	HFpEF (23)	*P*
Aortic flow parameters				
AO Forward Flow, ml	97±35	70±18*	70±19	0.02
AO Forward Flow indexed, ml/m^2	48±14	38±15*	36±10	0.01
AO Backward Flow, ml	2±1	1±4	2±3	0.86
AO Backward Flow indexed, ml	1±1	1±2	1±1	0.93
AO Max Area, mm^2	7±2	9±3	10±5	0.01
AO Min Area, mm^2	5±1	7±2*	8±4	**<0.001**
Flow Displacement systolic average,%	8±4	16±5*	23±10#	**<0.001**
Rotational Angle, °	0±0	-3±16	16±48#	0.04
Systolic Forward Flow, ml	90±37	66±18	77±27	0.15
Systolic Retrograde Flow, ml	2±1	4±4*	7±6#	**<0.001**
Systolic Flow Reversal Ratio,%	2±2	7±6*	11±10#	**<0.001**
Pulse Wave Velocity, m/s	4±2	11±8*	8±3	0.03

Data were represented as median ± IQR (%). *AO* aorta, *HCs* healthy cohorts, *HFpEF* heart failure with preserved ejection fraction. *P<0.05 compared young HCs versus old HCs; #P<0.05 compared old HCs versus HFpEF.

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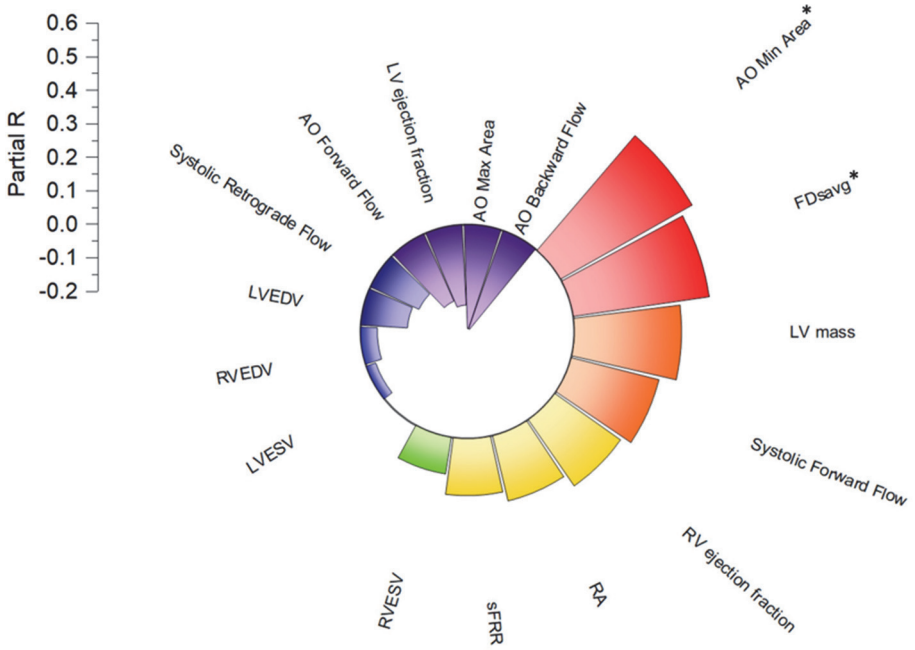
analysed average systolic aortic flow displacement (FDsavg), systolic flow reversal ratio (sFRR) and pulse wave velocity (PWV). In a sub-group analysis, we compared old HCs versus age-gender-matched HFpEF (N=10).

Results Differences were significant in mean age (P<0.001) among young HCs (22.9±3.5 years), old HCs (60.5±10.2 years) and HFpEF patients (73.7±9.7 years). FDs_{avg}, sFRR and PWV varied significantly (P<0.001) in young HCs (8 ±4%, 2±2%, 4±2m/s), old HCs (16±5%, 7±6%, 11±8m/s), and HFpEF patients (23±10%, 11±10%, 8±3). No significant PWV differences existed between old HCs and HFpEF. **(table 1, figure 1 and figure 2)**

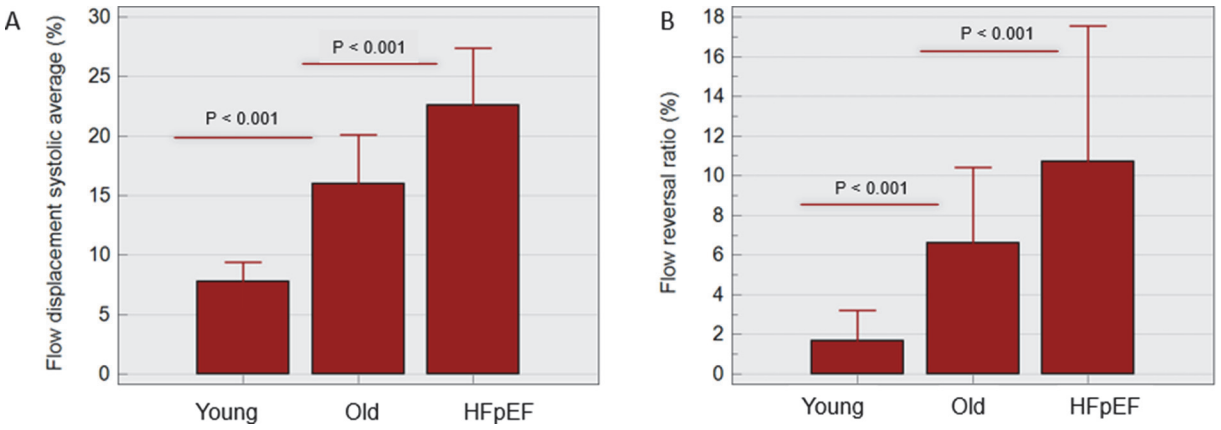
HFpEF had significantly higher FDs_{avg}versus old HCs (23 ±10% vs 16±5%, P<0.001). A FDsavg > 17.7% achieved 74% sensitivity, 70% specificity for differentiating them. sFRR was notably higher in HFpEF (11±10% vs 7±6%, P<0.001). A sFRR > 7.3% yielded 78% sensitivity, 70% specificity in differentiating these groups. **(figure 2)**

In sub-group analysis, FDs_{avg} remained distinctly elevated in HFpEF (22.4±9.7% vs 16±4.9%, P=0.029). FDs_{avg} of >16% showed 100% sensitivity and 70% specificity (P=0.01). Similarly, sFRR remained significantly higher in HFpEF (11.3 ±9.5% vs 6.6±6.4%, P=0.007). A sFRR of >7.2% showed 100% sensitivity and 60% specificity (P<0.001). **(figure 3)**

Discussion This study is one of the first to show a rise in sFRR and FDsavg in both ageing and HFpEF with distinct differences between the two groups even when matched for age and gender. CMR-derived FDsavg and sFRR can assist in early detection and sub phenotyping of HFpEF. Our recent work[2] demonstrated that these aortic flow abnormalities, particularly, FDsavg, can led to reduced exercise capacity and identify high risk individuals.



Abstract 31 Figure 1 Circular bar plot illustrating a linear correlation of age with left ventricular and aortic flow parameters via the multiple regression method using all other CMR variables as covariates - highlighting the correlation with FDsavg and AOmin



Abstract 31 Figure 2 **A** – Bar charts demonstrating flow displacement systolic average trends in young vs old healthy cohorts vs HFpEF patients. **B** – Bar charts demonstrating systolic flow reversal ratio trends in young and old healthy cohorts and HFpEF patients

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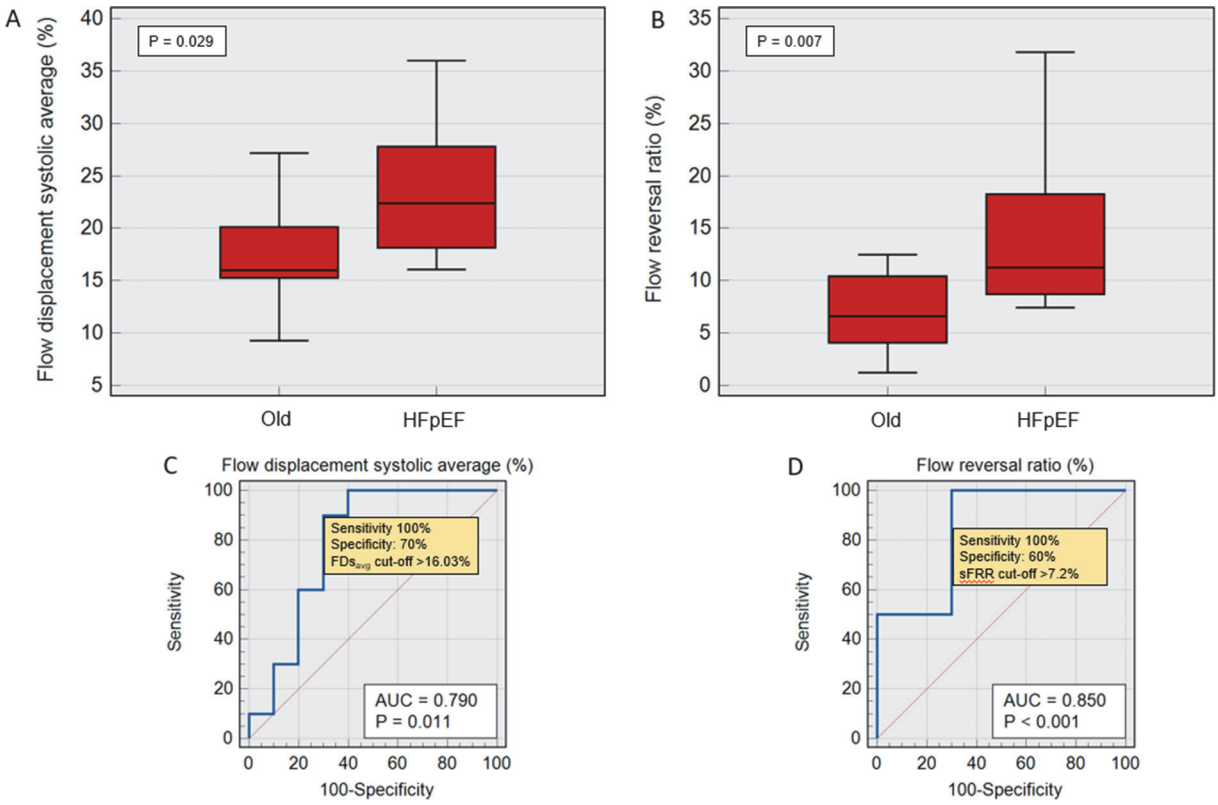

Abstract 31 Figure 3 **A** – Bar charts demonstrating average systolic flow displacement trends in age-gender-matched old healthy cohorts and HFpEF patients. **B** – Bar charts demonstrating systolic flow reversal ratio trends in age-gender-matched old healthy cohorts vs HFpEF patients. **C** and **D** – Receiver operating characteristic (ROC) with area under the curve demonstrating acceptable correlation in age-gender-matched old HCs vs patients with HFpEF

Conclusion Aortic flow haemodynamics (FDs_{avg} and sFRR) are significantly affected in ageing and HFpEF patients. Studies with larger and diverse cohort are required to draw definitive conclusions.

Acknowledgements

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32 SEX-SPECIFIC CARDIAC MAGNETIC RESONANCE PULMONARY CAPILLARY WEDGE PRESSURE MODEL PREDICTS OUTCOMES IN HEART FAILURE: A MULTI-CENTRE STUDY

[1]Ciaran Grafton-Clarke*, [1,2]Gareth Matthews, [1]Rui Li, [1,2]Hosamadin Assadi, [3]Peter Swoboda, [1,2]Chris Sawh, [1,2]Vassilios S Vassiliou, [4]Andrew J Swift , [1,2]Pankaj Garg. [1]*Norwich Medical School, University of East Anglia, Norwich, UK*; [2]*Norfolk and Norwich University NHS Trust, Norwich, UK*; [3]*University of Leeds, Leeds, UK*; [4]*University of Sheffield, Sheffield, UK*

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Introduction Non-invasive imaging is routinely used to estimate left ventricular (LV) filling pressure (LVFP) in heart failure (HF). Cardiovascular magnetic resonance (CMR) is emerging as an important imaging tool for sub-phenotyping HF and estimating pulmonary capillary wedge pressure (PCWP), a surrogate for LVFP.[1]

This research sought to investigate the following questions:

1) Does sex (male/female) influence CMR-derived PCWP?

2) Compared to a previously validated generic CMR-derived PCWP model,[1] does the use of a sex-specific model improve precision?

3) Is a novel sex-specific CMR-modelled PCWP prognostically significant?

Materials and Methods A derivation cohort (Norwich) involving patients with suspected HF underwent invasive right heart catheterisation and CMR within 24 hours of each other (*figure 1A*). This cohort was used to develop a sex-specific CMR model to estimate PCWP. The sex-specific CMR-derived PCWP model was developed using stepwise multivariate regression, using CMR variables significantly associated with PCWP during multivariate linear regression.

A validation cohort (Sheffield) of patients with confirmed heart failure underwent CMR (*figure 1B*) and was used to evaluate for the primary end-points of heart failure hospitalisation and MACE (composite cardiovascular death, HF hospitalisation, non- fatal stroke and myocardial infarction). A comparison in the prognostic performance between the generic model and the novel sex-specific CMR-derived PCWP models was performed.

Results 835 subjects were recruited from the Sheffield centre (derivation), and 454 subjects were recruited from the Leeds centre (validation).

Sex (male/female) does influence CMR-derived PCWP, since left atrial volume (LAV) and left ventricular mass (LVM) are higher in men (*figure 1C*). A novel sex- specific model incorporating LAV and LVM was created:

Sex-specific PCWP = 5.8 + (0.075*LAV) + (0.05*LVM) – (1.99*sex) [F=0; M=1]

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